

RE 497/597 A: Real Estate Data Modeling (Au25)**Instructor:** Shiqi Ding (shiqidi@uw.edu)**Classroom:** Gould Hall 007F**Class Days and Time:** Tuesdays and Thursdays 2:00 – 3:50 PM**Office Hours:** Thursdays 3:50 – 5:00 PM, or Mondays 11:00 AM – 12:00 PM by appointment

Course Description: This course will cover spatial analysis of real estate. The course will teach students methods in quantitative analysis of spatial data and how to collect, process, analyze, manage, and model a variety of data sets to make real estate decisions. Suitability and exploratory data analyses will be covered.

Course Objectives: There are three main objectives in this course: 1) examine real estate problems from a quantitative perspective; 2) describe and analyze data and variable relationships using a variety of spatial techniques; and 3) interpret and communicate the results to various audiences.

Textbooks and Readings: There are no required textbooks for the course. All the materials for the course are available online and can be accessed via Canvas.

Project-Based Course: This is a project-based course where students will conduct a suitability analysis. Esri applications and tools such as **ArcGIS Pro** and **Community Analyst** will be used to develop the suitability model. The purpose of the project is to learn how to use data for decision-making of real property acquisition by identifying the best locations for real estate assets. Students will form teams of three or four (to be decided) to work on the project and assignments. The project process involves gathering data, learning about different data sets and data types, processing and preparing data for analyses, understanding the data with basic descriptive statistics and spatial analysis, standardizing and transforming data, and weighting and combining the data to build a suitability model.

Course Structure:

By 1:30 PM Before Class	Complete discussion board (T) or module quiz (Th)
2:00 PM – 2:10 PM	Module and Project Q & A session (attendance highly encouraged)
2:10 PM – 3:00 PM	Discuss and demo group assignment (attendance required for assignment grade)
3:00 PM – 3:50 PM	In-class group assignment (attendance required for assignment grade)
After Class	Complete all readings, web courses, and module slides for next module

Course Requirements: The grade for the course is based on quizzes, discussion board, group assignments, and a final presentation.

- 1. Quizzes:** There is a short 15-question quiz for every week (due by Thursdays) that covers the required readings, web courses, and module slides. The quiz must be taken on Canvas, and you will be given 30 minutes to complete the quiz. The quiz should be taken after you have completed reviewing the resources for modules in that week. The quiz must be taken in one sitting, and you must complete the quiz on your own. You may not share the quiz questions and answers. There are no make-up quizzes allowed since you may take the quiz at any time between modules and only the highest 6 quiz scores will be counted toward your grade.
- 2. Discussion Board:** Students will participate in weekly discussion board activities that explore 1-2 open-ended questions related to course topics. Discussion board posts are due before class (on Tuesdays) each week as indicated in the course schedule. If you use AI tools to explore ideas or develop your thoughts for discussion posts, you must include a declaration of AI use at the end of your post, specifying which tool was used and for what purpose. All posts must reflect your own critical analysis and insights.

- 3. Group Assignments:** Students will work on group assignments in class. Each assignment builds on a component of the project. The instructions for the group assignments will start promptly at 2:10 PM. If you arrive after 2:10 PM or miss class, or leave before your group finishes the assignment, you will receive zero points for the assignment. Students will be able to complete several assignments during class, but the assignments aren't due until a later date to provide more time out of class to complete the assignment if needed.
- 4. Final Poster and Presentation:** Plan to prepare a poster and a 10-minute talk about your suitability model at the end of the course (7 minutes for presentation and 3 minutes for Q&A). The poster and talk should include: an overview of your project and project criteria, describe the data and variables used in your model and explain why, discuss the transformation of your data and any weights applied, and discuss the different sensitivity tests performed. Also, identify the top 5 sites selected and present one in detail. The grading criteria for the presentation and project are outlined in the rubric.

Important: All team members are responsible for preparing the poster. For groups that include both undergraduate and graduate students, graduate students will take the lead on presenting the poster, though all members should be prepared to answer questions about the project. Your poster should be in PDF format, 24" x 36", using the UW research poster templates:

<https://www.washington.edu/brand/templates/presentations/>

Q&A responsibility: Each group is required to ask at least one question throughout all presentations (to any group they choose, but not to every group). All students are responsible for answering questions from other groups. Any student that is not present for the presentation day will automatically have 10 points deducted from their final grade. Each group should submit the poster to Canvas as a PDF file by **5:00 PM on Tuesday, December 9th**.

Grading Rubric for Suitability Model Final Project and Presentation:

Criteria	Exemplary (A)	Acceptable (B)	Fair (C)	Unacceptable
Poster Content and Presentation of the Model	18 or 20 Points The model is demonstrated so the audience can easily understand its purpose and use and includes all the project components	16 Points The model is demonstrated so the audience can understand its purpose and use but more information is needed to understand the full scope	14 Points The model is demonstrated but there are some components regarding the purpose and use that are unclear or missing	12 or Less Points The presentation of the model lacks clarity, and its purpose and use are unclear, or components are missing
Poster Design	5 Points The overall design and layout of the poster is appealing, the maps and data are easy to read and understand with a color/pattern scheme and labels that are appropriate	4 Points Effort is seen in the overall design and layout of the poster, but some improvement is needed to make the overall design more appealing	3 Points The overall design and layout of the poster needs improvement to make the maps and data easier to read and changes to labels and color/pattern scheme are needed	2 or Less Points The overall design and layout are poor and there is no discernible design theme
Organization and Writing	5 Points The model is well organized with no grammar or spelling errors	4 Points The model is organized but there are a few writing errors	3 Points Little discernible organization and several writing errors	2 or Less Points No organization and significant writing errors
Peer-	5 Points	4 Points	3 Points	0 Points

Questions	Asked at least one insightful question that demonstrated a good understanding of the project or invited a discussion.	Asked one relevant question that demonstrated a good understanding of the project.	Asked one question that is unclear or irrelevant.	Asked no questions.
Time	5 Points Presentation is given in 7 minutes	4 Points Presentation is over/under by 2 minutes	3 Points Presentation is over/under by 3 minutes	2 or Less Points Presentation is over/under by more than 5 minutes

Course Evaluation:

	Number of Points	Total Number	Total Points
Quizzes	15	6	90 points
Discussion Board	10	9	90 points
Group Assignments	12	15	180 points
Poster and Lightning Talk	40	1	40 points
		Total Points:	400 points

Late Policy:

- 1. Quizzes, Discussion Board, and Final Presentation:** No late submissions will be accepted.
- 2. Group Assignments:** All assignments are due by the posted deadline (typically one day before the next class). Groups are expected to submit their best work by due date. In recognition that group projects often benefit from refinement, groups may request permission to resubmit an improved version within one week of the due date. All assignments must be submitted **by 5:00 PM of the last day of class (Thursday, December 4th)**.

Policy on Use of AI:

Given the hands-on, technical nature of this course using ArcGIS Pro, AI tools (such as ChatGPT) are encouraged for troubleshooting software issues, understanding GIS concepts, and problem-solving related to your suitability modeling project. However, AI tools are strictly prohibited during quizzes. For discussion board posts, if you use AI to explore open-ended questions, you must include a declaration of AI use, specifying which tool was used and for what purpose. Students or teams who use AI tools to assist with coursework or problem-solving must quote and cite any AI-generated text or materials (just as when using material from another source). You remain fully responsible for the accuracy, clarity, consistency, and coherence of all submitted work, as well as properly citing sources and avoiding plagiarism, regardless of whether AI tools are used. A Declaration of AI Use must be completed as part of each submitted deliverable when AI assistance is utilized.

Gradebook:

The gradebook will be updated weekly in Canvas (usually on Mondays). Please refer to the gradebook frequently to verify the number of points you have accumulated and your current standing in the course. Please notify me immediately if you believe an error has been made.

Contingency Plan:

If there is an unexpected disruption during the quarter that results in a university mandate of canceled classes and/or a truncated quarter, your grade will be calculated based on the total points you have earned out of the total points possible up to the date of closure. Therefore, it is important to stay on track for all coursework and not depend on future assignments to improve your grade. This policy will only go into effect if there is a mandated university-wide cancellation of the remainder of the quarter.

Undergraduate Grading Scale:

A	4.0	94% and above	376+ points
	3.9	93%	372 – 375 points
A-	3.8	92%	368 – 371 points
	3.7	91%	364 – 367 points
	3.6	90%	360 – 363 points
	3.5	89%	356 – 359 points
B+	3.4	88%	352 – 355 points
	3.3	87%	348 – 351 points
	3.2	86%	344 – 347 points
B	3.1	85%	340 – 343 points
	3.0	84%	336 – 339 points
	2.9	83%	332 – 335 points
B-	2.8	82%	328 – 331 points
	2.7	81%	324 – 327 points
	2.6	80%	320 – 323 points
	2.5	79%	316 – 319 points
C+	2.4	78%	312 – 315 points
	2.3	77%	308 – 311 points
	2.2	76%	304 – 307 points
C	2.1	75%	300 – 303 points
	2.0	74%	296 – 299 points
	1.9	73%	292 – 295 points
C-	1.8	72%	288 – 291 points
	1.7	71%	284 – 287 points
	1.6	70%	280 – 283 points
	1.5	69%	276 – 279 points
D+	1.4	68%	272 – 275 points
	1.3	67%	268 – 271 points
	1.2	66%	264 – 267 points
D	1.1	65%	260 – 263 points
	1.0	64%	256 – 259 points
	0.9	63%	252 – 255 points
D-	0.8	62%	248 – 251 points
	0.7	61%	244 – 247 points
E	0.0	60% and below	243 points and below

Graduate Grading Scale:

A	4.0	94% and above	376+ points
	3.9	93%	372 – 375 points
A-	3.8	92%	368 – 371 points
	3.7	91%	364 – 367 points
	3.6	90%	360 – 363 points
	3.5	89%	356 – 359 points
B+	3.4	88%	352 – 355 points
	3.3	87%	348 – 351 points
	3.2	86%	344 – 347 points
	3.1	85%	340 – 343 points

B	3.0	84%	336 – 339 points
	2.9	83%	332 – 335 points
B-	2.8	82%	328 – 331 points
	2.7	81%	324 – 327 points
	2.6	80%	320 – 323 points
	2.5	79%	316 – 319 points
C+	2.4	78%	312 – 315 points
	2.3	77%	308 – 311 points
	2.2	76%	304 – 307 points
	2.1	75%	300 – 303 points
C	2.0	74%	296 – 299 points
	1.9	73%	292 – 295 points
C-	1.8	72%	288 – 291 points
	1.7	71%	284 – 287 points
E	1.6	70% and below	283 points and below

Class Attendance Policy:

Attending class in person is mandatory and is strongly encouraged. You will only benefit from the discussions and work on the assignments if you are present for the in-person sessions.

Classroom Etiquette:

1. Be punctual for all classes.
2. Be respectful of the beliefs of others.
3. Turn off cell phones and other electronic devices that may be disruptive to class.
4. If you need to leave the room, be as quiet as possible when entering and exiting the class while in session.
5. The use of any recording device without permission is prohibited.

Meeting Protocol for the Virtual Classroom (if necessary):

The class will be taught in person at Gould 007F. This section is only applicable if the course moves online during the quarter. Zoom is integrated into Canvas and you should be able to connect to each class via Canvas. Please test all technology before class, including your video, audio, and Wi-Fi connection. Make sure your camera, mic, and speakers are all working properly. Choose an appropriate space for joining the virtual classroom (a quiet place where you can focus and will not be disturbed). If there is a technical issue with Zoom, please watch the Canvas announcements as I will post instructions on how we will handle the technical issue.

During class, please follow these guidelines:

- Muting and unmuting your mic: Your microphone is set on mute when you enter the virtual classroom. Please unmute yourself to speak. Mute your mic when you are not speaking to eliminate ambient feedback and other unwanted background noises.
- Start Video: Your video is off by default at the start of the class. Please start your video when class starts and keep the camera on during the entire class, especially when you are speaking. For class engagement, it is important that we can see each other. If you are interrupted during class or need to take a quick break to use the restroom, please turn off your video but restart it as soon as you return.
- Chat: We will not be using this function in this course since it can be distracting. If you have questions or comments, please unmute your mic or raise your hand.
- Raise Hand: During class, you can use the Raise Hand button to speak.
- Share Screen: When presenting using a PowerPoint or another application, share the application only rather than your desktop to protect your privacy.

Academic Integrity: Academic integrity is an important value of this course. Students engaging in academic misconduct, particularly plagiarism, will be referred to the Dean's office for further investigation. VeriCite is used in Canvas for all written assignments and plagiarism will not be tolerated in this course. If you use more than 5 words in a row from another document without quotation marks, even if you include the source in parentheses, this is considered plagiarism. Exact language must be in quotes and there are very few situations where quotes are acceptable. I must report plagiarism and it will result in an investigation that can lead to disciplinary sanctions. Please read the [UW Student Conduct Code](#) with special attention to the [Academic Misconduct section](#).

Disability Accommodations: The Runstad Department of Real Estate is committed to ensuring learning opportunities for students with disabilities. If you would like to request academic accommodations due to a disability, please contact the [Office of Disability Resources for Students](#). If you have a letter from the Office of Disability Resources for Students indicating you have a disability that requires academic accommodations, please present it to me as soon as possible so we can discuss specific accommodations for this course.

Religious Accommodations Policy: Washington state law requires that UW develop a policy for accommodation of student absences or significant hardship due to reasons of faith or conscience, or for organized religious activities. The UW's policy, including more information about how to request an accommodation, is available at [Religious Accommodations Policy](#). Accommodations must be requested within the first two weeks of this course using the [Religious Accommodations Request form](#).

Student Athletes: Student athletes who have conflicts with attending any lecture or submitting any assignment should submit a list of such conflicts along with a letter from the athletics department by **Tuesday, September 30, 2025**. I will reasonably accommodate any student athlete who follows this guideline.

Safety: Students are advised to refer to UW policies and procedures to ensure their safety and security on campus. For more information, go to: <http://www.washington.edu/safecampus/>. To report threats, seek advice, or get counseling, dial 206-685-SAFE (7233).

ArcGIS Online: Each student will be given access to ArcGIS Online. ArcGIS Online can only be used for academic purposes related to your course work. For all Esri products, you will sign in **through the website** and use the **ENTERPRISE LOGIN**. The ArcGIS organization's URL is [uwsupplemental.maps.arcgis.com](https://www.esri.com/arcgis/products/arcgisonline/enterprise). You will then continue to sign in using your UW NetID and UW Password. At the end of the quarter, your ArcGIS Online account will be deleted as well as any materials that you have stored in your online account. If you wish to have your account active until you graduate, you must notify me in writing before the end of the quarter. If you request extended access, your account and all related materials will be deleted when you graduate. You can move content in your account to other accounts using the ArcGIS Online Assistant: <https://ago-assistant.esri.com/>. At the end of the quarter please delete all your groups and content.

ArcGIS Pro: Each student will be given access to the ArcGIS Pro software. ArcGIS Pro can only be used for academic purposes related to your course work. **Important:** Using some geoprocessing tools in ArcGIS Pro and Esri Community Analyst may incur credit usage against your AGOL account.

Suzzallo Library GIS Lab: The Suzzallo Library has Geospatial Data Resources and a GIS lab. The lab also has staff that can provide assistance with basic GIS tasks. Visit the following webpage for the hours that the GIS assistant is available: <https://guides.lib.uw.edu/research/gis/lab>

Esri Resources:ArcGIS Online and ArcGIS Pro

ArcGIS Online Sign In (<https://www.arcgis.com/home/signin.html>) ArcGIS

Online Help (<https://doc.arcgis.com/en/arcgis-online/>)

ArcGIS Online Credits (<http://www.esri.com/software/arcgis/arcgisonline/credits>)

ArcGIS Pro Help (<https://pro.arcgis.com/en/pro-app/help/main/welcome-to-the-arcgis-pro-app-help.htm>)

Learning Resources

Esri Training (<https://www.esri.com/training/>)

Applied Analysis (<http://desktop.arcgis.com/en/analytics/casestudies/>)

GeoNet The ESRI Community (<https://community.esri.com/>)

Learn ArcGIS (<https://learn.arcgis.com/en/>)

ArcGIS for Developers (<https://developers.arcgis.com/>)

Esri User Conference (<https://www.esri.com/en-us/about/events/uc/overview>)

Schedule:

Class and Due Dates	Topics
Week 0 Thursday September 25, 2025 Introduction:	Welcome and Syllabus Overview ArcGIS Pro/ArcGIS Online Account Setup and Group Folder Setup
Week 1 Tuesday September 30, 2025 Discussion Board 1: Module 1: Module 2: Assignment 1&2:	Due by 1:00 PM Introduction to Suitability Modeling Getting Started with ArcGIS Pro Group Assignment 1&2 due by October 1, 2025, at 11:59 PM
Week 1 Thursday October 2, 2025 Quiz 1: Module 3: Assignment 3:	Due by 1:00 PM Intro to Spatial Data, Geodatabases, and Data Management Group Assignment 3 due by October 7, 2025, at 11:59 PM
Week 2 Tuesday October 7, 2025 Discussion Board 2: Module 4: Assignment 4:	Due by 1:00 PM Datasets and Data Sources Group Assignment 4 due by October 8, 2025, at 11:59 PM
Week 2 Thursday October 9, 2025 Quiz 2: Module 5: Assignment 5:	Due by 1:00 PM Geoprocessing and Editing Group Assignment 5 due by October 13, 2025, at 11:59 PM
Week 3 Tuesday October 14, 2025 Discussion Board 3: Module 6: Assignment 6:	Due by 1:00 PM Coordinate Systems and Projections Group Assignment 6 due by October 15, 2025, at 11:59 PM
Week 3 Thursday October 16, 2025 Quiz 3: Module 7: Assignment 7:	Due by 1:00 PM Geocoding and Creating XY Event Layers Group Assignment 7 due by October 20, 2025, at 11:59 PM
Week 4 Tuesday October 21, 2025 Discussion Board 4: Module 8: Assignment 8:	Due by 1:00 PM Spatial and Table Joins Group Assignment 8 due by October 22, 2025, at 11:59 PM
Week 4 Thursday October 23, 2025	No Class. The instructor is attending the 2025 ACSP conference.

Week 5 Tuesday October 28, 2025 Discussion Board 5: Module 9: Descriptive Statistics Assignment 9:	Due by 1:00 PM Group Assignment 9 due by October 29, 2025, at 11:59 PM
Week 5 Thursday October 30, 2025 Quiz 5: Module 10: Correlation Assignment 10:	Due by 1:00 PM Group Assignment 10 due by November 3, 2025, at 11:59 PM
Week 6 Tuesday November 4, 2025 Discussion Board 6: Module 11: Basics of Cartography Assignment 11:	Due by 1:00 PM Group Assignment 11 due by November 5, 2025, at 11:59 PM
Week 6 Thursday November 6, 2025 Quiz 6: Module 12: Analyzing Spatial Patterns Assignment 12:	Due by 1:00 PM Group Assignment 12 due by November 10, 2025, at 11:59 PM
Week 7 Tuesday November 11, 2025	No Class. Veterans Day. Classes are not in session on university holidays.
Week 7 Thursday November 13, 2025 Quiz 7: Module 13: Identifying Clusters Assignment 13:	Due by 1:00 PM Group Assignment 13 due by November 17, 2025, at 11:59 PM
Week 8 Tuesday November 18, 2025 Discussion Board 8: Module 14: Standardizing and Transforming Data Assignment 14:	Due by 1:00 PM Group Assignment 14 due by November 19, 2025, at 11:59 PM
Week 8 Thursday November 20, 2025 Quiz 8: Module 15: Weight Data and Create Index Assignment 15:	Due by 1:00 PM Group Assignment 15 due by November 24, 2025, at 11:59 PM
Week 9 Tuesday November 25, 2025 Discussion Board 9: Module 16: Performing Sensitivity Analysis Assignment 16:	Due by 1:00 PM Group Assignment 16 due by November 26, 2025, at 11:59 PM
Week 9 Thursday November 27, 2025	No Class. Thanksgiving. Classes are not in session on university holidays.

Week 10 Tuesday December 2, 2025 Discussion Board 10: Module 17: Assignment 17:	Due by 1:00 PM Overlays and Locating Top Sites No Group Assignment.
Week 10 Thursday December 4, 2025	Final Posters Presentation PDF due by December 9, 2025, at 5:00 PM